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How Well Does the FDA FAERS Reporting System Work for “Grey Area” Substances?

The FDA is responsible for regulating products that make up 25% of the U.S. economy: food, drugs, cosmetics, supplements, medical devices, tobacco products, and more. The FDA’s FAERS adverse event system is powered by multiple algorithms and produces an interactive dashboard that consumers and medical professionals alike can use. The “signals” that FAERS produces are a key part of the FDA and DEA’s regulatory decision-making toolkit, helping determine when to regulate, issue warnings about, or ban certain products.

But when a legal, herbal supplement like kratom, which has gained popularity in recent years as an opioid alternative, enters the system, how well does this algorithm function? The FAERS dashboard uses a combination of algorithms and statistical models to create “signals” about a drug and its corresponding adverse event reports. The safety signals tell the FDA that there’s a common adverse event coming from a specific drug or combination of drugs that needs to be looked at more deeply, and may require regulation or warning labels to prevent consumer harm.

The central algorithm this system uses is the Multi-item Gamma Poisson Shrinker (MGPS), invented by Dr. William DuMouchel at Oracle in 1999. MGPS is a disproportionality algorithm that handles skewed reporting in adverse event data, and sets “thresholds” for these drug+event pairs to help make sense of thousands of drug reports. A key part of how FAERS knows when to create a signal for a certain drug, MGPS prevents outlier or rare drug-event pairs from creating such a signal. If a drug-event pair happens often enough and meets MGPS thresholds, the dashboard will create a signal that alerts the FDA of concerning health outcomes based on its statistical correlation algorithm.

My investigation is about how FAERS handles “grey area” substances like kratom, a legal and highly addictive drug derived from a South Asian plant that has recently been synthesized and isolated into popular products available in gas stations, smoke shops, and grocery stores nationwide.

I spoke with Oliver Grundmann, a key kratom researcher at the University of Florida’s School of Pharmacy, who has used FAERS to inform his research on kratom’s pharmacology and harmful effects on consumers.

“9% of US adults have used or are using kratom in the U.S., based on our latest prevalence estimate. That’s 25 to 30 million people. The problem with the FDA’s FAERS reporting system is that many of these extremely harmful adverse reactions come from combining kratom with other, typically innocuous substances. FAERS’ MGPS algorithm is not necessarily prepared to deal with the combination of kratom reporting biases: consumers feel shame associated with this ‘gas station heroin’ addiction, which leads patients to not disclose their dependence on it to their doctors. These patient experiences are being left out of HCP case reports, which should detail the adverse events that come from kratom use,” says Grundmann.

“Also, in the rare case that consumers themselves use the FAERS dashboard, they often leave out important information. We need to know what other substances they used that day that could precipitate an adverse event. A healthcare provider may be able to ask these questions and report them, but lay-people leave out poly-drug interactions that may play a part in these FAERS signals.”

FAERS relies on accurate case reports from both healthcare providers and consumers. Without adequate knowledge from the consumer community and full disclosures from patients to their healthcare providers, FAERS’ MGPS system may be falling short in supporting the FDA’s regulatory goals. This new crop of legal drugs presents a unique problem for the algorithms in place used by the FDA and DEA.

In the current landscape of healthcare utilization companies and their solutions, data scientist Rajat Goel, recently laid off from his position at healthcare consulting nonprofit IPRO, says these models may soon be far out of date.

“We’re looking at an outdated healthcare algorithm landscape. So many higher-ups in healthcare companies are hesitant to buy-in to AI solutions for their internal management systems. But the reality is, their competitors have already started to contract with private companies that optimize things like proprietary adverse event+drug signaling models and predictive prescription models,” says Goel.

“Frankly, MGPS is out of date now that Artificial Intelligence is in the picture. Healthcare AI models nowadays are coming up with more efficient systems for looking at substances like kratom and other legal, yet-to-be-regulated drugs,” says Goel.

The publication QuarterWatch, which was created to regularly report on the limitations of the FAERS system, was discontinued by the Institute for Safe Medication Practices years ago, according to a spokesperson at ISMP. Right now, there are no independent organizations reviewing the FAERS system or its core MGPS algorithm.

In my future efforts to investigate the MGPS algorithm and FAERS system, I’ll need to speak more with kratom experts who have experience with consumers using kratom products. Goel says that when MGPS was created, there were far fewer products that existed in the post-regulatory market. These drugs include those like kratom, whose isolates like 7-hydroxymitraginine and mitragynine pseudoindoxyl are just reaching the illegal import market and look more like Sackler-era OxyContin than any other drug before them.

“A big part of this outdated algorithm problem has a lot to do with what the algorithm is trained on. When there’s a substance that the regulatory bodies don’t have a name for yet, it’s hard for consumers and HCPs alike to record them in the FAERS system,” Goel says. Currently, the

FAERS dashboard has two inputs for kratom-derived products: “kratom/herbals” and “7-hydroxymitragynine.” But more products are being created from the kratom leaf and their derivative origins are difficult to parse from unregulated packaging.

“I think this legal sale and illegal import market is quickly outpacing algorithms that are just not prepared for such drugs,” says Grundmann.

As I continue investigating the kratom landscape and the FAERS system designed to regulate such products, I’ll speak to employees at the DEA and FDA about how FAERS is used currently. I’ll also speak to university researchers and experts on the MGPS algorithm and Bayesian statistical models, how they’re being used today, the limitations of these models in the current regulatory environment, and what improved algorithms are in development. I also hope to speak with current FDA employees about the use of AI in development of new signaling systems.